Robust Sequential Learning in Random Order Networks

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ABSTRACT

In the sequential learning problem, agents in a network attempt to predict a binary ground truth, informed by both a noisy private signal and the predictions of neighboring agents before them. It is well known that social learning in this setting can be highly fragile: small changes to the ordering, network topology, or even the strength of the agents' private signals can prevent a network from converging to the truth. We study networks that achieve *random-order asymptotic truth learning*, in which almost all agents learn the ground truth even when the decision ordering is selected uniformly at random. We analyze the robustness of these networks, showing that those achieving random-order asymptotic truth learning are resilient to a bounded number of adversarial modifications. We characterize necessary conditions for such networks to succeed in this setting and introduce several graph constructions that learn through different mechanisms. Finally, we present a randomized polynomial-time algorithm that transforms an arbitrary network into one achieving random-order learning using minimal edge or vertex modifications, with provable approximation guarantees. Our results reveal structural properties of networks that achieve random-order learning and provide algorithmic tools for designing robust social networks.

KEYWORDS

Social networks, Social Learning, Sequential learning, Bayesian learning, Robust networks, Approximation algorithms, Submodular Optimization

ACM Reference Format:
William Guo, Edward Xiong, and Jie Gao. 2026. Robust Sequential Learning in Random Order Networks. In *Proc. of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026), Paphos, Cyprus, May 25 – 29, 2026*, IFAAMAS, 15 pages.

1 INTRODUCTION

The ability of a population to correctly aggregate dispersed information is fundamental to effective societal decision-making. Whether predicting a market crash, identifying a public health threat, or creating new social behaviors, success hinges on how weak signals can compound into stronger signals, and how these signals propagate through the social network. It is in the interest of both individual agents and network designers to understand the circumstances enabling society to learn as a collective, particularly when many network parameters—including the network topology and decision ordering—can be subject to change.

Proc. of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026), C. Amato, L. Dennis, V. Mascardi, J. Thangarajah (eds.), May 25 – 29, 2026, Paphos, Cyprus.

Decision learning in a social setting, that is, social learning, is a well-studied topic in economics and social networks [7, 14, 21]. In the classical model of sequential learning [6, 7, 10, 29, 32], agents take turns to make decisions, and each agent sees a private signal with uncertainty as well as the previous agent's decisions. One major concern in this setting is *herding* or *information cascade*: with constant probability, a few early agents make a wrong decision and subsequent agents ignore their own private signals, conforming to the actions of the agents they observe. This results in the majority of the network converging to the wrong value, even with perfectly rational Bayesian agents.

In this work, we consider sequential learning on a social network, following the model adopted in a few recent papers [4, 5, 20, 30]. A set of agents make predictions sequentially about an underlying binary ground truth. Each agent uses Bayesian inference to compute the truth value with the maximum likelihood, using their independent bounded private signal as well as their neighbor's predictions to construct their own belief. Recent work on this model shows that the quality of the information spread throughout a network is dependent on its graph structure, as well as the order in which individuals act. The ideal learning outcome is characterized by the notion of *asymptotic learning*, in which all but $o(n)$ agents in a network of size n correctly learn the ground truth with probability approaching 1 as n goes to infinity. However, the precise relationship between graph topology/decision ordering and whether a network is capable of learning asymptotically still remains elusive.

Of particular interest are networks that achieve asymptotic learning even when the decision ordering is chosen uniformly at random. In practice, imposing a carefully crafted strategic ordering on a network can be challenging when there is no global coordination. Random ordering presents the average or expected behavior. Thus, a network that supports asymptotic learning with a random ordering is particularly appealing. Such networks have inherent structural properties preserved under most orderings, allowing us to isolate the role of topology on social learning.

However, due to the inherent difficulty of preserving key graph structures under random orderings, networks that learn under random orderings are more difficult to design. In the literature, only two instances of networks learning under random-orderings have been identified in the literature: the "celebrity network" in [5], and symmetric expander graphs satisfying the "local learning requirement" in [4]. Both networks seemingly possess graph properties robust to adversarial deletion: [4] explicitly shows the existence of a network obtaining random-order learning, even after the removal of all but an arbitrary constant fraction of the network.

In this paper, we focus on characterizing and enabling asymptotic learning with random ordering. We characterize the robustness properties of random-order learning networks to adversarial modifications. Furthermore, we offer constructions for networks achieving random-order learning through different means, and provide an algorithmic approach to enable arbitrary networks to learn using a small number of modifications.

1.1 Our Contributions

To motivate our exposition into random-order learning networks, we first show that asymptotic learning under a strategic ordering can be fragile and breaks down with small changes. We exhibit a graph family that achieves asymptotic learning under a carefully crafted strategic ordering only when the uncertainty of the agent's private signals is in a specified range. This demonstrates the lack of stability of networks that rely on such strategic orderings discussed in [20] and [30]. In contrast, we show that asymptotic learning under random ordering is robust to a constant number of adversarial modifications, generalizing the robustness properties discussed in [4] to arbitrary random-order learning networks. Namely, a random-order learning network with learning rate $1 - \varepsilon$ is robust to $o(\frac{1}{\varepsilon})$ many modifications. This bound is nearly worst-case tight – we show that removing $\Theta(\frac{1}{\varepsilon})$ many vertices from a learning celebrity network can result in a non-learning network.

Motivated by the robustness property, we provide constructions and algorithmic approaches to design networks that support random-order learning. We start with the classical example of a complete network that suffers from an information cascade [6]. We first show that any learning network must have an independent set of size $\omega(1)$, a property that a complete graph does not have. Surprisingly, by introducing a superconstant number of "guinea pigs" (degree 1 vertices), we find that the complete network can be "boosted" to achieve random-order asymptotic learning. In fact, this boosting approach is not limited to using guinea pigs: any graph structure capable of achieving asymptotic learning under a strategic order can be embedded into the complete graph to enable random-order learning.

While these constructions are specific to the complete graph, one may wonder how to boost an arbitrary network family to achieve random-order learning using a minimum number of edge/vertex modifications. We introduce a randomized algorithm to do exactly this by greedily boosting vertices with high coverage of other non-learning vertices. Our approach is closely related to the influence maximization framework introduced in [19], and more broadly submodular maximization [26, 33], which we use to show that our algorithm is a $O(g(n) \log n)$-approximation of the minimum number of edge modifications needed to achieve random-order learning, for any superconstant function $g(n) = \omega(1)$.

Our algorithm, however, assumes the existence of an efficient "learning oracle" which determines whether a vertex achieves a high learning rate. The construction of such an oracle is a highly non-trivial problem and relates to various other learning problems that have been shown to be computationally hard. We conjecture the hardness of computing such an oracle for general graphs. Meanwhile, we give a deterministic linear-time heuristic to check if an agent achieves asymptotic learning, and discuss when this heuristic is useful.

1.2 Related Work

There is a large variation in model and parameter choices in the social learning setting. We focus on the work most relevant to ours, and additionally refer the readers to some surveys in the field [2, 14, 21, 24].

Enabling Social Learning: To avoid herding in sequential learning, previous work suggests limiting social interactions, for example removing the visibility between agents early in the sequence [27, 28], or introducing a stochastic social network among the agents [1]. A recent work also studied the timing effects of strategic behaviors of agents to delay actions [16]. When the private signals are unbounded (i.e., with probability arbitrarily close to 1), asymptotic learning occurs almost certainly [15, 29].

Sequential vs. Repeated learning: We study sequential learning where each agent makes one prediction following a linear order. In the repeated learning setting, agents broadcast a binary decision at every new time step t [22, 23]. For Bayesian agents with repeated learning, asymptotic truth learning is achieved if the graph is undirected and the distribution of private beliefs is not atomic[1] [23], or if agents repeatedly share with others their current beliefs [24].

Hardness of Bayesian Learning: It is well known that general Bayesian belief inference is difficult [11]. In the repeated social learning setting, it has been shown that computing the optimal decision is NP-hard even at $t = 2$ [18]. Generally, inferring the most likely ground truth value is PSPACE-hard [17]. In the sequential setting, determining whether a particular network can obtain a high learning rate under some strategic ordering is also NP-hard [30].

Influence Maximization: Our algorithm for enabling random-order learning relies on finding a set of highly influential agents, which is a problem dating back to [12]. Prior work establishes approximation algorithms for maximizing expected influence under various diffusion models [8, 19]. Notably, so long as the influence objective is submodular, one can immediately obtain strong approximation guarantees [26, 33].

Robustness of Social Learning: Existing literature predominantly analyzes the robustness of social learning in light of faulty or misguided decision rules. For example, Bohren [3] considers a setting where some agents blindly predict based on their private signal, and shows that significantly over/underestimating the proportion of uninformed agents leads to undesirable behavior. Mueller-Frank [25] concerns the repeated learning setting, noting that the standard weighted average (DeGroot) decision model is not robust to arbitrarily small adjustments made to a single agent at each iteration. In the sequential learning setting, Arieli et. al [4] observe that networks learning under random orderings are robust to random vertex deletions. In particular, from the perspective of any agent not deleted, removing a large subset of vertices selected uniformly at random equates to the agent arriving early in the decision ordering. In this sense, learning under random vertex deletions is embedded in the definition of uniform random orderings.

2 SETTING

We define a *network* $\mathcal{F}$ to be an infinite family of undirected graphs with unbounded order $\sup_{G \in \mathcal{F}} |V(G)| = \infty$. For ease of notation,

[1]Note that a binary Bernoulli distribution for private signal as in our setting is not non-atomic.

throughout this paper we describe networks as sequences $\mathcal{F} = \{G_n\}_{n\in\mathbb{N}}$ of undirected graphs $G_n = \{V_n, E_n\}$ where $|V_n| = n$. Each graph G_n represents n agents who with perfect knowledge of the entire graph topology. Each graph is equipped with a decision ordering (equivalently, permutation) $\sigma_n : [n] \to [n]$, where σ_n is a bijection mapping each agent to a unique index. We distinguish between the *strategic ordering* setting, in which one may carefully choose the agent's ordering for any graph G_n, and the *random ordering* setting, in which σ_n is chosen uniformly at random from the $n!$ possibilities. In either case, the decision ordering is known by all agents in advance.

Each agent seeks to learn the ground truth signal $\theta \in \{0, 1\}$, where $\Pr(\theta = 0) = \Pr(\theta = 1) = \frac{1}{2}$. Each agent is given a random private signal $p_v \in \{0, 1\}$, representing an independent noisy measurement of the ground truth. The agent's private signals are correlated with θ with a common probability $q > 1/2$.

As per the decision ordering σ_n, agents will go in sequence to make their predictions. For any agent v, let $N(v)$ denote the set of neighbors of v in a graph. In sequential learning on a graph G, v can see the actions of their neighbors arriving before them in the ordering, namely in the set $N_\sigma(v) = \{u \in V_n : (u, v) \in E_n, \sigma(u) < \sigma(v)\}$. Each agent v then announces a prediction $a_v \in \{0, 1\}$ visible only to each of its neighbors succeeding it in the ordering. The ordering σ induces an acyclic directed graph $\vec{G}$ which orients each edge in G to be pointing from the lower index to the higher index in σ. An agent v's *subnetwork* $B(v)$ is defined on the set of all $u \in \vec{G}$ for which a u-v directed path exists in $\vec{G}$. Such vertices are called the ancestors of v. Informally, it is the set of all agents that may ultimately influence v's belief and action.

Following the literature in social learning [6], we assume all agents are perfectly Bayesian, and will predict the most probable ground truth value given their knowledge of G, σ, q, their private noisy signal, and the predictions of their neighbors arriving before them. For simplicity, if an agent finds that it is equally likely for $\theta = 0$ or 1, the agent will side with their own private signal, although our results generalize to arbitrary tie-breaking rules.

For each agent $v \in V$, its *learning rate* $\ell_\sigma(v) := \ell_\sigma(v, q)$ given a fixed ordering σ is the probability that v correctly predicts the ground truth: $\ell_\sigma(v) = \Pr(a_v = \theta)$. The learning rate of a graph $G = \{V, E\}$ paired with a fixed ordering σ is the expected fraction of agents predicting the ground truth, i.e.

$$L_\sigma(G) = \frac{1}{|V|} \sum_{v \in V} \ell_\sigma(v).$$

Similarly, we define the *random-order learning rate* of an agent v to be $\ell(v) = \mathbb{E}_\sigma[\ell_\sigma(v)]$, averaging the learning rate over all possible orderings σ weighted uniformly. For an event $A \subseteq S_n$ dependent only on the random ordering of the vertices (where S_n is the set of all possible permutations of $[n]$), the random-order learning rate conditioned on A is denoted as $\ell(v \mid A) = \Pr(a_v = \theta \mid \sigma \in A)$. The random-order learning rate of a graph G is the expected fraction of agents predicting the ground truth given a decision ordering selected uniformly at random, or

$$L(G) = \mathbb{E}_\sigma[L_\sigma(G)] = \frac{1}{|V|} \sum_{v \in V} \ell(v).$$

The learning rate of a network $\mathcal{F} = \{G_n\}_{n\in\mathbb{N}}$ is the *asymptotic learning rate* of its graph family, or $\lim_{n\to\infty} L(G_n)$. We say that a network obtains *random-order asymptotic truth learning* if $L(G_n) \to 1$ as $n \to \infty$. When $\mathcal{F}$ is equipped with a sequence of fixed orderings $\{\sigma_n\}$, we say the network achieves *strategic order learning* if $\lim_{n\to\infty} L_{\sigma_n}(G_n) = 1$. We will abbreviate random order (resp. strategic order) asymptotic truth learning as simply random-order (resp. strategic order) learning, or even just "learning". An agent v with learning rate approaching 1 as n goes to infinity *learns* or *achieves learning*.

We now describe a few helpful results with proofs in Appendix A. The first is the generalized improvement principle, an extension of Proposition 1 in [20]:

Proposition 2.1 (Generalized improvement principle). *Given a graph $G = \{V, E\}$ and a particular vertex $v \in V$, consider a subgraph $G' = \{V, E'\}$ of G such that $E' \subseteq E$, and for all $u, w \neq v$, $(u, w) \in E \implies (u, w) \in E'$. Then for any fixed ordering σ, the learning rate of v in graph G, denoted by $\ell_\sigma(v)$, is at least that of the learning rate of v in G' denoted by $\ell'_\sigma(v)$. Further, under random orderings $\ell(v) \geq \ell'(v)$.*

In essence, the generalized improvement principle ensures that each agent does at least as well as if it could only see a subset of its neighbors. This is true due to the rationality (Bayesian inference) of each agent, along with their knowledge of the graph G and ordering σ. As a corollary, one can see that each agent learns at least as well as its neighbors before it in the decision ordering; inductively, this implies that each agent does at least as well as all previous agents in its subnetwork.

Next, we establish a general property of random-order learning in a network setting, which holds independently of the network's decision rules.

Lemma 2.2. *For a vertex v in G which obtains a random learning rate at least $\ell(v) \geq 1 - \varepsilon$, and an event $A \subseteq S_n$ which occurs with probability $\Pr(A) = \frac{|A|}{n!}$ on uniform random orderings, the learning rate of v conditioned on A is bounded by*

$$\ell(v \mid A) \geq 1 - \frac{\varepsilon}{\Pr(A)}.$$

Finally, we note any vertex with many learning neighbors itself must achieve learning.

Lemma 2.3. *For any network $\mathcal{F} = \{G_n\}_{n\in\mathbb{N}}$ and vertex $v = \{v_n : v_n \in G_n, n \in \mathbb{N}\}$, if v has $d = \omega(1)$ neighbors that achieve learning, then v itself also achieves learning.*

One may expect that as long as the random ordering places v after one of its d learning neighbors, it too will learn. However, each neighbor has a lower learning rate conditioned on being early in the ordering, as, in expectation, they have less information to inform their beliefs (see Lemma 3.2). If v arrives too early, none of its neighbors before it may be in a position that enables learning. The decline in learning rate conditioned on being early in an ordering is a recurring obstacle throughout our analyses.

Remark. *Many of our results invoke asymptotic notation despite referencing a single instance of a graph or vertex. As these results generalize to arbitrary n, we find it acceptable to do this for brevity, as is* *standard for theoretical findings. However, we find it difficult to justify guarantees involving "asymptotic learning" for single graph/vertex instances. As in Lemma 2.3, we will occasionally clarify our results with reference to infinite families/sequences as opposed to fixed instances.*

3 ROBUSTNESS IN NETWORK LEARNING

3.1 Fragility of Strategic-Order Learning

To motivate our exposition into random-order learning networks, we first highlight the fragility of social learning under strategic orderings. Prior work (e.g., [20]) studied a myriad of networks that learn under strategic orderings. This type of learning is heavily dependent on the strategic ordering, yet even so, it can be fragile to other network parameters. Here, we introduce a network family that learns conditionally depending on q (the probability of private signals being correct), even after fixing the underlying graph topology and strategic ordering on the vertices.

Our construction is shown in Figure 1, in which there is a special vertex v with four neighbors arriving before it, and a set of k neighbors $\{w_i\}$ arriving after it. Each w_i also has an independent neighbor that arrives before it. All the agents $\{w_i\}$ feed their predictions into a common neighbor u_0 arriving later. We let $k = \log n$, though any superconstant value of k suffices. All remaining vertices are arranged in a chain following the vertex u_0.

In essence, the ability of the network to learn depends solely on whether u_0 is a learner. If u_0 could observe all (or even a constant fraction) of the independent private signals from the agents $\{w_i\}$, it would be able to correctly determine the ground truth with high probability by a simple Chernoff bound. However, all agents w_i observe the action of v. For $q > q_0$ for some $q_0 \approx 0.7887$, each w_i will default to the action of v, whereas for $q < q_0$, each will act on its independent private signal with constant probability. In the former case, all w_i's will simply regurgitate the action of v, which u_0 will also adopt. But as v is not a learner, neither is u_0. In the latter case, u_0 sees enough private signals that it is able to learn with high probability. The detailed proof can be found in Appendix B.

Proposition 3.1. *There exists a strategic-order network $\mathcal{F}$ that obtains asymptotic learning for $q < q_0$ and does not learn for $q > q_0$, for some fixed threshold q_0.*

This example is rather counter-intuitive in that the network stops learning when the quality of private signals *improves*. However, this is not universal: a similar construction of a network that only learns for *high* values of q can be found in Appendix B. It can be seen that a small modification to the network, e.g., inserting or removing one of the early-arriving neighbors of v, can influence the entire network's learning behavior via a similar analysis.

A fundamental strength of random-order learning networks is the ability to learn almost independently of the decision ordering imposed on it. Such networks are able to learn seemingly due to inherent properties of the underlying topology. A natural question is whether such properties can be broken via minor perturbations to the network structure. In the next section, we investigate the robustness of random-order learning networks against *adversarial* edge/vertex modifications.

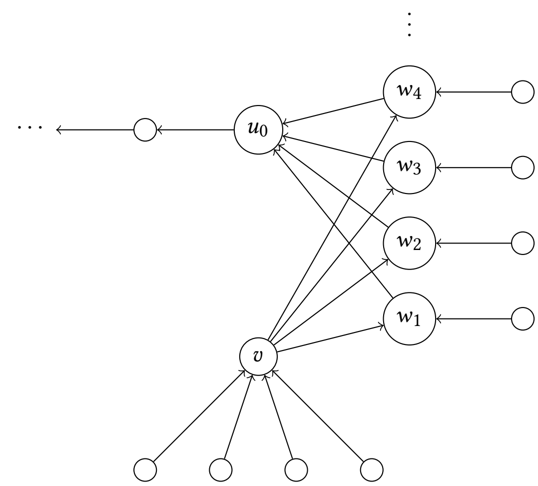


Figure 1: A network that obtains learning for $q < q_0$ but not for $q > q_0$, for some $q_0 \in [\frac{1}{2}, 1]$. The edges in the graph are oriented to indicate the direction of information flow.

3.2 Robustness of Random-Order Learning

In this section, we argue that a network's robustness to edge/vertex insertion/deletions can be made explicit in terms of its random-order learning rate. We start with a rather intuitive observation: conditioned on being early in the decision ordering, an agent does worse than when conditioned on being later in the ordering. This can be attributed to the generalized improvement principle, as an agent arriving early and only seeing a few of its neighbors will be less informed than an agent arriving later and seeing more of its neighbors.

Lemma 3.2 (Monotonicity). *Given a graph G, fix an agent $v \in G$. For all $i \in [n]$, let A_i be the event that v is at index i, or $\sigma(v) = i$. Then $\Pr(a_v = \theta \mid A_i) \leq \Pr(a_v = \theta \mid A_{i+1})$ for all $1 \leq i \leq n-1$.*

A simple proof can be found in Appendix A. Using monotonicity, we can show that the random-order learning error of a graph only increases linearly in the number of modifications. The essence of the proof relies on the probability of any given agent arriving earlier in the random ordering before any of the vertex modifications.

Theorem 3.3. *Given a graph G and a vertex $v \in G$ with random-order learning rate $1 - \varepsilon$, after k vertex deletions, v obtains learning rate at least $1 - (k+1)\varepsilon$.*

Proof. Let D be the set of k deleted vertices, and let G' be the graph obtained by deleting D from G. For $n = |G|$, the size of G' is $n - k$. Consider the probability distribution $\mathbb{P}_v$ of orderings on G', obtained by first uniformly at random picking an ordering σ of G such that v precedes all $w \in D$, and then removing all the $w \in D$ to obtain an ordering σ' on G'.

We use $\ell(v \mid \mathbb{P}_v)$ to denote the learning rate of $v \in G'$ under this distribution of orderings. Similarly, we use A_v to denote the event that v precedes all vertices $w \in D$ in the original graph G.

The probability $\Pr(A_v)$ under a uniformly random ordering of G is $\frac{1}{k+1}$, so by Lemma 2.2 we have

$$\ell(v \mid A_v) \geq 1 - (k+1)\varepsilon.$$

For any ordering σ such that A_v occurs, we can trivially remove all vertices in D to obtain an ordering σ' of G'. The learning rate of v under σ and σ' is the same, since v is not affected by any vertex that arrives after it does. Therefore,

$$\ell(v \mid \mathbb{P}_v) = \ell(v \mid A_v) \geq 1 - (k+1)\varepsilon.$$

Now observe that in the distribution $\mathbb{P}_v$, the probability that v is found at index i is

$$\Pr(\sigma(i) = v \mid \mathbb{P}_v) = \frac{\binom{n-i}{k}}{\binom{n}{k+1}}.$$

and for the orderings such that $\sigma(i) = v$, all other vertices are uniformly random in the remaining indices. Importantly, note that $\Pr(\sigma(i) = v \mid \mathbb{P}_v)$ is monotonically decreasing in i. Thus by Lemma 3.2, the learning rate of v in G' is

$$\begin{aligned}\ell(v) &= \frac{1}{n-k}\sum_{i=1}^{n-k} \ell(v \mid \sigma(i) = v) \geq \sum_{i=1}^{n-k} \frac{\binom{n-i}{k}}{\binom{n}{k+1}} \ell(v \mid \sigma(i) = v) \\ &= \ell(v \mid \mathbb{P}_v) \geq 1 - (k+1)\varepsilon.\end{aligned}$$

as desired. □

The same approach works to prove the following analogous statements on other modifications.

Corollary 3.3.1. *Given a graph G and a vertex $v \in G$ with random-order learning rate $1-\varepsilon$, after k vertex insertions, v obtains learning rate at least $1-(k+1)\varepsilon$.*

Corollary 3.3.2. *Given a graph G and a vertex $v \in G$ with random-order learning rate $1-\varepsilon$, after k edge deletions or insertions, v obtains learning rate at least $1-\frac{2k+1}{2}\varepsilon$.*

In this context, we take care to note that a vertex insertion allows for adding an arbitrary set of edges from the new vertex to any pre-existing vertices in the graph. Thus, we find that networks achieving random asymptotic learning are robust against a number of modifications dependent on the learning rate of the network:

Theorem 3.4. *Given a network $\mathcal{F}$ with random-order asymptotic learning rate $1-\varepsilon$, consider the network $\mathcal{F}'$ where each $G'_n \in \mathcal{F}'$ is obtained from some $G_m \in \mathcal{F}$ after at most k modifications, which may be vertex deletions, vertex insertions, edge deletions, and edge insertions. Then, $\mathcal{F}'$ obtains random-order learning rate at least $1-(k+1)\varepsilon$. If $\varepsilon = o(1)$ and $k = o(1/\varepsilon)$, then $\mathcal{F}'$ also achieves asymptotic learning.*

We observe that for general random-order learning networks, this bound is nearly tight. As an example, consider the celebrity network [5], which consists of complete bipartite graphs $K_{n-k,k}$ with $k = o(n)$ and $k = \omega(1)$.[2] The larger partition of $n-k$ vertices can be thought of as "commoners", while the smaller partition of k vertices represents "celebrities". In a given random ordering, with high probability, the first celebrity in the ordering observes the

[2]When $k = \Omega(n)$ or $O(1)$, a poorly informed herd occurs with constant probability. For example, when $k = \Omega(n)$, with constant probability, the first celebrity sees only a constant number of commoners, so its probability of failure is at least constant.

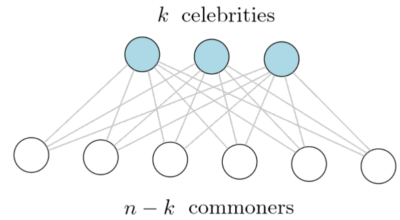


Figure 2: Celebrity graph, as described in [5]

independent signals of many commoners. Upon aggregating this information into a single, high-quality signal, the celebrity can disseminate it to the rest of the network. When $k = \Omega(\log n)$, the learning rate of the celebrity network is at most $1-\Theta(1/k)$, as at least $\Theta(n/k)$ commoners arrive before the first celebrity with probability $1-o(1)$. Then, for $\varepsilon = \Theta(1/k)$, one may adversarially delete the $k = \Omega(1/\varepsilon)$ celebrities from the network, leaving $n-k$ lone vertices, which do not obtain asymptotic learning.

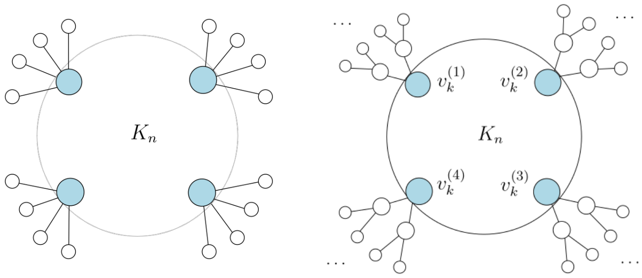
4 CONSTRUCTING ROBUST NETWORKS

4.1 Boosting a Complete Network

Prior to this work, the understanding of networks obtaining random-order learning has been quite limited, both in terms of necessary/sufficient properties and tangible examples of graph families. We first note a key necessary property of random-order learning networks, relying on a simple observation of Bayesian learning: the strength of a signal is dependent on the number of independent private signals backing it. It is known that each decision is a deterministic function of the announced private signals before it (see Lemma 4 in Appendix B of [20]). Then, for any agent v learning under random orderings, it is necessary that, with probability approaching 1, a superconstant number of agents preceding v act according to their own private signals. Conversely, if there is a constant probability that v's subnetwork contains only $O(1)$ ancestors acting on their own private signals, with constant probability all of those ancestors receive incorrect private signals, implying v has a learning rate bounded away from 1. We relate the number of agents predicting their own private signal to the size of the maximum independent set, and formalize a network-wide property in Proposition 4.1. The proof can be found in Appendix D.

Proposition 4.1. *Any network $\mathcal{F}$ obtaining random-order learning must have maximum independent set size $\omega(1)$.*

Now, we introduce a new type of graph family obtaining asymptotic truth learning under random orderings that relies primarily on constant-degree vertices. We contrast this with the celebrity graph, in which *all* agents learn with high probability: there are no peripheral agents that do not learn yet still contribute significantly to the underlying learning scheme. In our construction, the constant-degree vertices do not learn themselves, but serve as guinea pigs for a select number of special vertices that propagate the signal to the rest of the network. Ignoring the constant-degree vertices, the rest of the network forms a complete graph, which is known not to achieve learning [6].



(a) (b)

Figure 3: (a): By equipping a small number of vertices in a K_n graph with many guinea pigs, the entire network can be boosted to achieve learning. (b): K_n can also be boosted with $\sqrt{n}$ complete binary trees of height $\log\log\log n$ (total size $\log\log n$) to achieve random-order asymptotic learning.

Proposition 4.2. *There exists a network $\mathcal{F}$ obtaining random-order learning that no longer achieves learning when all vertices of degree 1 are removed from each graph $G_n \in \mathcal{F}$.*

An instance of such a network is shown in Figure 3a. In essence, $g(n)$ special vertices in the complete graph K_n are each equipped with $h(n)$ guinea pigs with degree 1, where $g(n), h(n) = \omega(1)$ and $g(n) \cdot h(n) = o(n)$ so the total number of added vertices is negligible. With probability approaching 1, each special vertex sees $\Omega(h(n))$ of its guinea pigs, which all give their own private signals. By a Chernoff bound, we may deduce that each special vertex learns, and so by Lemma 2.3, the entire K_n graph achieves learning.

Despite the complete graph having poor learning capabilities on its own, Proposition 4.2 implies that it can achieve learning upon "boosting" a small number of its vertices. One may note that equipping a vertex with a superconstant number of guinea pigs is not the only way to boost a vertex to achieve a high learning rate. Indeed, we show that *any* construction permitting a vertex to achieve high learning rates under fixed orderings can be embedded into a K_n network to boost the network at large. An example of a graph boosted by complete binary trees is shown in Figure 3b.

Formally, suppose we are given a network $\mathcal{F}' = \{G'_n\}$ equipped with strategic orderings $\{\sigma_n\}$ such that for all $k \geq 1$, imposing the ordering $\sigma_k = \{v_1, v_2, \cdots, v_k\}$ on G'_k permits v_k to learn, i.e. $\ell_{\sigma_k}(v_k) = 1 - o(1)$. Now, for sufficiently large n, let $k = \log\log n$. We begin with the complete graph K_n, and pick $\sqrt{n}$ distinct vertices, denoted $S = \{v_k^{(i)} : i \in [\sqrt{n}]\}$. For each vertex $v_k^{(i)}$, attach a copy of $G'_k \setminus v_k$ so the induced graph on $\{G'_k \setminus v_k\} \cup v_k^{(i)} = G'_k$. For $m = n + \sqrt{n}(\log\log n - 1)$, we set G_m to be the final result, and allow $\mathcal{F} = \{G_m\}$ to be our final random-order learning network.

Proposition 4.3 (Embedding arbitrary learning structures into random-order learning networks). *Given a network $\mathcal{F}' = \{G'_n\}$ and strategic orderings $\{\sigma_n\}$ for which $v = \{v_n : v_n \in G'_n\}$ obtains asymptotic learning, the construction described above yields a network $\mathcal{F}$ that achieves random-order asymptotic learning.*

A complete proof of both Proposition 4.2 and Proposition 4.3 can be found in Appendix D. As a proof sketch for the latter, we note the probability over the random ordering that the agents in any G'_k copy are ordered identically to the sequence given by σ_k is

$$\frac{1}{k!} \geq \frac{1}{(\log\log n)^{\log\log n}} \geq \frac{1}{n^\delta}$$

for any $\delta > 0$. For any copy of the graph G'_k for which this occurs, the corresponding $v_k^{(i)}$ achieves learning by the Proposition 2.1, i.e., its learning rate is at least that if it could only see its neighbors in the copy of G'_k. As the copies of G'_k are disjoint, we conclude each $v_k^{(i)}$ has a probability $\geq 1/n^\delta$ of achieving learning. There are $\sqrt{n}$ disjoint copies of $v_k^{(i)}$, so we expect many to achieve learning.

The key technical challenge arises from the fact that the earlier copies of $v_k^{(i)}$ have an exponentially lower probability of learning, as necessarily all other vertices in its corresponding G'_k copy must appear before it. To circumvent this, we condition on the number of graphs whose entire vertex set appears before an index $j = n/\omega(1)$ for some small superconstant denominator. Since the internal ordering of each graph copy is independent, we proceed as above to show that at least one vertex $v_k^{(i)}$ before index j achieves learning. Thus, any vertex u after index j learns, and as each agent's learning rate monotonically increases in its index, u achieves learning overall.

We find this result rather surprising, as the inherent difficulty of preserving key graph properties under random orderings would seem to suggest that complex learning structures (e.g., binary trees) are not necessary for random-order learning. Notably, each boosted vertex v_i has a low probability of learning on its own, but it is almost guaranteed that in aggregate, at least one v_i learns.

4.2 Algorithmically Boosting Networks

We now generalize boosting to arbitrary graphs in order to enable networks to achieve random-order learning while minimizing the number of modifications (limited to edge insertions/deletions and vertex insertions). [3] Our algorithm runs in polynomial time, taking a fixed graph as input. To obtain the asymptotic guarantees for an entire network, one would need to execute our algorithm on each $G_n \in \mathcal{F}$. The proofs in this section can be found in Appendix C.

A key assumption we make is that the algorithm has access to a learning oracle that can determine whether a vertex learns above a certain threshold or not.

Definition 4.1. *Let $f(v, t) : V \times (0, 1) \to \{0, 1\}$ be an oracle that outputs 1 if v learns at rate at least t under random orderings, and 0 otherwise.*

To our best knowledge, it is unclear whether such a function can be computed efficiently. We discuss this and useful approaches to simulating a learning oracle in Section 4.3.

When modifying a graph, the number of new vertices should be sublinear in the number of vertices in the original graph so as not to overrule the existing network. If the number of vertices that need to be boosted is $\Omega(n)$, as is the case in a sparse graph with constant average degree, we would like to avoid adding a separate set of vertices for each one of them. Instead, we replicate the structure of the celebrity graph by connecting each target vertex with a superconstant number of learning vertices, allowing us to

[3] Our analysis excludes vertex deletions, as events on the induced subgraph of the original vertices may change non-uniformly under deletions.

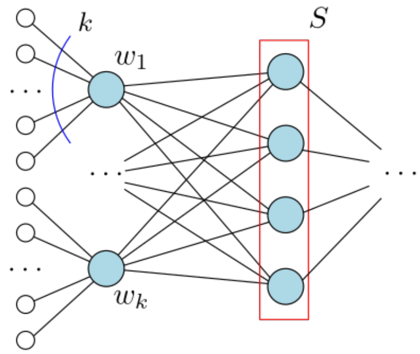


Figure 4: BoostAgents boosts the vertices $S \subseteq V$ by connecting them to an added set of celebrities $\{w_1, \cdots, w_k\}$.

'celebrity-ify' vertices in a selected set. This subroutine is described in BoostAgents.

Procedure 1 BoostAgents (G, S, k)

Construct $T = \{z_{11}, \ldots, z_{kk}\} \cup \{w_1, \ldots, w_k\}$
Add edges (z_{ij}, w_i) for all $i, j \in [k]$
For each $v \in S$ and $i \in [k]$, add edge (w_i, v)

As depicted in Figure 4, we introduce a set of k "celebrities" $\{w_i\}$ that are each equipped with independent guinea pigs to guarantee their learning rate is high. By Lemma 2.3, the added celebrities are capable of boosting the learning quality of the set S.

Claim 4.4. *Suppose $k = \omega(1)$. After BoostAgents on a set $S_n \subseteq V_n$, as $n \to \infty$, each $v \in S_n$ achieves random-order asymptotic truth learning.*

The number of vertices added by BoostAgents is $k^2 + k = \omega(1)$. Since k can be any arbitrary slow-growing function, the number of new vertices can be made to be within the optimum by an additive value that is slightly superconstant. However, the number of edges added is $|S|k + k^2$. While k^2 is relatively small, as Theorem 3.4 suggests that non-learning networks require $\omega(1)$ modifications to achieve learning, $|S|$ may be as large as $\Omega(n)$, such as in sparse networks with constant average degree.

We instead focus on selecting a set S that is of *near-optimal* size. Suppose for a network G, there exist some modifications to make it a learning network. For any single non-learning agent, with a constant probability, it must be reachable by at least 1 such modification in its subnetwork for its decision to be changed. In other words, the set of vertices that are modified/incident on a modification must reach all but $o(n)$ non-learners with constant probability, and this provides a lower bound on the minimum number of modifications required to achieve learning. As a matter of fact, calling BoostAgents on a set of vertices maximizing coverage of non-learners is indeed a suitable algorithmic approach. In Lemma 4.5, we first argue that any non-learner reachable by many boosted vertices will also receive the effects of boosting.

Lemma 4.5. *For a set $S \subseteq V$ on which BoostAgents has been called, suppose that each $u \in S$ has been equipped with $1/\varepsilon = \omega(1)$ many guinea pigs. Then any vertex v that is reachable from S w.p. $1 - \delta$ over the ordering has learning rate $\geq 1 - \delta - 2\sqrt[3]{81\varepsilon}$. In particular, the probability v is reachable by some $u \in S$ for which $\sigma(u) \geq n\varepsilon\alpha$ is at least $1 - \delta - \alpha\varepsilon$, for any $\alpha = \omega(1)$, $\alpha = o(\frac{1}{\varepsilon})$.*

The key challenge to proving Lemma 4.5 is again monotonicity: in this context, we would like to select vertices with high coverage, but coverage increases the earlier a vertex is in the ordering. A reasonable worry is that our algorithm will select vertices that only have high coverage when placed too early to be affected by the celebrities added by BoostAgents, thereby avoiding the effects of boosting. Fortunately, this is not the case: as stated in the second half of the lemma, coverage tends to be fairly linear. That is, moving a vertex earlier by εn indices can only decrease the probability it reaches some other vertex by at most ε.

Using Lemma 4.5, Lemma 4.6 follows immediately, suggesting the optimal number of modifications through boosting alone is within an arbitrarily slow-growing factor of the true optimum. As networks achieving asymptotic learning may leave $o(n)$ vertices uncovered, the exact optimum for our purposes may not be well defined. For precision, we fix a tolerance parameter $T := T(n) = o(n)$ and mandate that in expectation over the random ordering, at most T vertices are unaffected by the modifications. All our bounds hold for any threshold T.

Lemma 4.6. *Given a network $\mathcal{F} = \{G_n\}_n$, consider any $M := M(n)$ such that it is possible for $\mathcal{F}$ to obtain random-order learning by performing at most $M(n)$ vertex insertions or edge insertions/deletions to each G_n, where in expectation at most $T := T(n)$ non-learning vertices are unaffected by the modifications. Then there exists some $M_B := M_B(k, n)$ such that $M_B \leq O(k)M + 2k^2$ and it is possible to achieve random-order learning by adding at most $M_B(k, n)$ edges/vertices to each G_n solely through calls to BoostAgents(G, S, k), with at most T unaffected vertices in expectation.*

It remains to develop an algorithm to find a set S that maximizes its coverage. We leverage results from optimization problems with submodular functions [19, 26, 33]. Notably, fixing the size of the set $|S| = m$, the maximum coverage by the best m-element set can be approximated to within a constant factor by simply greedily picking the agent yielding the largest marginal increase at each iteration. Equivalently, fixing some target number of non-learners to be covered, the number of vertices to be boosted, chosen by the greedy approach is at most a $O(\log n)$ factor greater than the optimum. Our full algorithm can be seen in Algorithm 1.

We first show that coverage as a function of a set S is submodular. Let $V' \subseteq V$ be the set of non-learners in G, and $C_\sigma(S)$ denote the number of agents in V' reachable from S under ordering σ. Then, $C = \frac{1}{n!}\sum_\sigma C_\sigma(S)$ is the expected number of agents reachable from S under a uniformly random ordering.

Definition 4.2. *[19] Given some domain space U, a submodular function $f : 2^U \to \mathbb{R}$ satisfies*

$$f(S \cup \{x\}) - f(S) \geq f(T \cup \{x\}) - f(T)$$

for all $S \subseteq T \subseteq U$ and $x \in U$.

Claim 4.7. *For any $V' \subseteq V$, $C : 2^{V'} \to \mathbb{R}$ defined above is nonnegative, monotone, & submodular.*

Computing $C(S)$ exactly requires iterating through $n!$ many possible orderings, which is not computationally feasible. However, with randomization, we may use Monte Carlo sampling to estimate $\tilde{C}(S) = \frac{1}{N}\sum_{i=1}^{N} C_{\sigma_i}(S)$, for sampled orderings $\{\sigma_1, \dots, \sigma_N\}$. Iteratively, we append the vertex v that maximizes $\tilde{C}(S \cup \{v\})$. For sufficiently large N, $\tilde{C}(S)$ can be made within an additive $\frac{1}{n}$ of $C(S)$ at all times. After accounting for the additive error, we may use results from the submodular optimization literature to show that our randomized greedy algorithm obtains a $\log n$-approximation.

Algorithm 1 BoostGraph-MonteCarlo(k)

$S \leftarrow \emptyset$, $\varepsilon = \frac{1}{n^2}$, $\delta = \frac{1}{n^9}$, $N = \frac{n^2}{2\varepsilon^2}\ln\frac{2}{\delta}$
Using a learning oracle, determine the set of non-learners V'
while $\tilde{C}(S) < |V'| - T$ **do**
  Sample orderings $\{\sigma_1, \dots, \sigma_N\}$ uniformly at random
  For each $v \in V'$, compute $\tilde{C}(S \cup \{v\}) = \frac{1}{N}\sum_{i=1}^{N} C_{\sigma_i}(S \cup \{v\})$
  Let $v^* = \arg\max_{v \in V'} \tilde{C}(S \cup \{v\})$
  Set $S \leftarrow S \cup \{v\}$
end while
Run BoostAgents(G, S, k)

Lemma 4.8. *Let $ALG := ALG(k, n)$ be the number of edges added by Algorithm 1. Then for sufficiently small $k = O(\log n)$ and any $M := M(n)$ as defined in Lemma 4.6, $ALG \le O(k \log n) \cdot M$ with probability at least $1 - \frac{1}{n^8}$.*

The runtime of Algorithm 1 is polynomial, and by Lemma 4.5 it obtains learning as well. Taking $k = g(n)$ for any desired function $g(n) = \omega(1)$, we conclude with the following theorem:

Theorem 4.9. *Given efficient query access to a learning oracle, for any $k = \omega(1)$, $T = o(n)$, and network $\mathcal{F} = \{G_n\}$, Algorithm 1 can be run in polynomial time to modify each graph in $\mathcal{F}$ to achieve random-order learning such with probability $\ge 1 - \frac{1}{n^8}$, at most T agents are unaffected in expectation. Furthermore, the number of modifications made $O(g(n)\log n)$-approximates the optimum, for any function $g(n) = \omega(1)$.*

4.3 Efficiently Identifying Learning Vertices

To preface this section, we conjecture the hardness of computing, or even reasonably approximating any learning oracle. Even under a fixed ordering, computing the learning rate of a given vertex is difficult. In the repeated learning setting, [18] gives an exponential time algorithm for general networks, and also proves the NP-hardness of computing any individual agent's decision when the private signals may be asymmetrically assigned (i.e. $\Pr(a_v = 1 \mid \theta = 0) \neq \Pr(a_v = 1 \mid \theta = 1)$). Though their results are intended for the repeated learning setting, their key hardness result is directly applicable to a graph coupled with a strategic ordering in our setting. Given this, we suspect that it is difficult to determine whether an arbitrary vertex $v_n \in G_n$ achieves a learning rate above some threshold t is difficult, and hence weeding out which vertices achieve learning is generally not computationally feasible.

Conjecture 4.10. *For general graphs G, $v \in V$, and $t \in [0, 1]$, computing the output of a learning oracle $f(v, t)$ is NP-hard.*

We may instead attempt to identify simple graph structures that permit learning. Heuristically, many learning networks rely on agents that learn due to a specific graph structure independent of the rest of the network, which we refer to as "first learners". For random-order learning, one indicator of whether a vertex is a first learner with high probability over the ordering is the number of independent neighbors it has. For example, a vertex boosted by BoostAgents becomes a first learner as many of its neighbors will feed the targeted vertex their own independent private signals.

We describe a simple deterministic method to check if a vertex is a first learner. In particular, one may simply compute $\sum_{u \in N(v_n)} \frac{1}{\deg(u)}$ in $O(\deg(v)) = O(n)$ time. In essence, this tests whether an agent can directly see many independent signals from its neighbors.

Claim 4.11. *For $v = \{v_n : v_n \in G_n\}$, if*

$$\sum_{u \in N(v)} \frac{1}{\deg(u)} = \omega(1)$$

then v learns under random orderings.

The key insight is similar to the proof of Proposition 4.1 in that for any $v \in G$, the expected number of neighbors arriving first within their own neighborhood is σ is $\sum_{u \in N(v)} \frac{1}{1+\deg(u)} = \Theta(\sum_{u \in N(v)} \frac{1}{\deg(u)})$. Each such vertex is forced to predict its own private signal. If v can see the private signals of a superconstant number of neighbors, it will be able to learn with probability approaching 1. A full proof can be found in Appendix D.

We observe that a graph may learn despite no agent having a high probability of being a first learner. The construction in Proposition 4.3 shows that even arbitrarily complex structures with low probabilities of staying intact under random orderings can be made essential for a network's learning. Under strategic orderings, there exist networks capable of learning without a clear set of first learners, such as the butterfly graph described in [20]. Unlike the heuristic in Claim 4.11, which can be efficiently used to test whether a vertex is a first learner, it is unclear how one may feasibly determine if a network learns via either of these alternative mechanisms.

5 CONCLUSION

In this work, we explored the structure of truth learning in social networks under uniformly random decision orderings. Our results highlight the fragility of networks learning under strategic orderings, while simultaneously proving the robustness of random-order learning networks. We also characterized various structural requirements, such as the necessity of large independent sets to achieve learning, and introduced constructions which demonstrate surprising capacity for learning under random orderings.

Our algorithm for transforming arbitrary networks into random-order learning networks uses a greedy approach to pick influential agents in the network to boost. With the use of an oracle that can calculate random-order learning rates of specific vertices, this algorithm runs in polynomial time and succeeds with high probability. Recall that [30] proves that the fixed-order learning decision problem is NP-hard. We conjecture that a similar result holds for the random-order learning rate of a network, though we speculate that a standard approach, such as reducing from 3SAT as done in both

[30], [11], is unlikely to work due to their reliance on a particular ordering of the belief announcements.

Future work may explore the computational complexity of computing and/or approximating learning rates, and the design of alternative algorithms that don't rely on such oracles. One direction may be to derive stronger relationships between a network's learning rate and properties of its underlying graph topology. To this end, it would be interesting to obtain stronger characterizations of learning networks, especially stronger sufficiency conditions on a network that would guarantee random-order asymptotic learning.

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A PROOFS OF HELPFUL RESULTS

Proof of Lemma 2.2. We can decompose the learning rate of v as

$$\ell(v) = \ell(v \mid A)\Pr(A) + \ell(v \mid \overline{A})\Pr(\overline{A}).$$

Trivially we have $\ell(v \mid \overline{A}) \leq 1$, and $\Pr(\overline{A}) = 1 - \Pr(A)$ so we have

$$\ell(v \mid A)\Pr(A) + 1 - \Pr(A) \geq \ell(v) \geq 1 - \varepsilon.$$

This simplifies to

$$\ell(v \mid A) \geq 1 - \frac{\varepsilon}{\Pr(A)}$$

as desired. □

To prove the general improvement principle (Proposition 2.1), we leverage a simpler claim: that each agent performs as least as well as its best neighbor.

Proposition A.1 (Proposition 1 in [20]). *Given a graph/ordering pair G, σ and a particular vertex $v \in V$, $\ell_\sigma(v) \geq \max\{\ell_\sigma(u) : u \in N(v), \sigma(u) < \sigma(v)\}$.*

Proof of Proposition 2.1. In G, construct v' to be a copy of v along with its incident edges in G', so $N(v') \subseteq N(v)$. Then, construct an ordering σ' on $G \cup \{v'\}$ by inserting v' immediately before v in σ. For sake of analysis, we now tweak the setting such that the private signals $p_{v'}, p_v$ are forced to be identical instead of independent. (This is different than conditioning on the event $p_{v'} = p_v$, which would give information to both agents. One may think of this modification as v, v' collectively receiving a single private signal.) As each v, v' still has private signal strength q, the subnetworks observed by v'/v are identical to those seen under σ, and v' doesn't affect the decision of v, we have that $\ell_\sigma(v) = \ell_{\sigma'}(v)$ and $\ell'_\sigma(v) = \ell_{\sigma'}(v')$.

Now, consider adding an edge between v' and v. As both v, v' are Bayesian and all information available to v' is also available to v, v does not change its decision due to the addition of this edge, so its learning rate remains unchanged. But then by Proposition A.1, $\ell_{\sigma'}(v) \geq \ell_{\sigma'}(v')$. Thus, $\ell_\sigma(v) \geq \ell'_\sigma(v)$. □

We may intuit that any agent with many learning neighbors itself must also learn. However, this isn't a straightforward conclusion. Conditioned on arriving before the agent itself, the learning rate of any given neighbor is necessarily lowered. This is particularly harmful even when many of the agent's neighbors arrive beforehand. We prove a lower bound on the learning rate of any vertex with many learning neighbors, incurring a $\log d$ multiplicative factor in the error of the agent.

Lemma A.2. *Any vertex v with at least d neighbors obtaining learning rates at least $1-\varepsilon$ will itself achieve a learning rate of at least*

$$\ell(v) \geq 1 - \frac{1}{d+1} - 2\varepsilon \ln d.$$

While it may seem counterintuitive that this bound weakens with the degree of the agent for large d, by Proposition 2.1, we may substitute any $d' \leq d$ into the above bound.

Proof. Let M be the set of neighbors $u \in N(v)$ such that $\ell(u) \geq 1-\varepsilon$. For any ordering σ of the vertices, let $M_\sigma \subseteq M$ contain all $u \in M$ such that u arrives before v. If there are d neighbors achieving learning $1-\varepsilon$, the probability v arrives in a certain position relative to the d neighbors is

$$\Pr(|M_\sigma| = k) = \frac{1}{d+1}$$

for all k such that $0 \leq k \leq d$. Now, we condition on the event $|M_\sigma| = k$. If $k=0$ then $\ell(v \mid M_\sigma = \emptyset) = q$ trivially. Otherwise, for a fixed vertex $u_0 \in M$, consider the event $A_{u_0,k}$ where u_0 is the kth vertex to arrive among all vertices in $M \cup \{v\}$ and v arrives after u_0. Also define $B_{u_0,k}$ as the event that one of $A_{u_0,1}, A_{u_0,2}, \dots, A_{u_0,k}$ occurs. We have

$$\Pr(A_{u_0,k}) = \frac{d+1-k}{d(d+1)},$$

and since all $A_{u_0,k}$ are disjoint for fixed u_0, this gives

$$\Pr(B_{u_0,k}) = \sum_{i=1}^{k} \Pr(A_{u_0,k}) = \frac{kd - k(k-1)/2}{d(d+1)}.$$

By Lemma 3.2, we also observe that for all $k' < k$, we have

$$\ell(u_0 \mid A_{u_0,k}) \geq \ell(u_0 \mid A_{u_0,k'}).$$

Thus we have $\ell(u_0 \mid A_{u_0,k}) \geq \ell(u_0 \mid B_{u_0,k})$. By Lemma 2.2, we have

$$\ell(u_0 \mid A_{u_0,k}) \geq \ell(u_0 \mid B_{u_0,k}) \geq 1 - \frac{\varepsilon}{\Pr(B_{u_0,k})}.$$

Now we relate this learning rate to the learning rate of v: define the event $C_{u_0,k}$ where u_0 is the kth vertex to arrive and v is the $k+1$th vertex to arrive among all vertices in $M \cup \{v\}$. Note that $C_{u_0,k}$ is a subset of $A_{u_0,k}$ and

$$\ell(u_0 \mid C_{u_0,k}) = \ell(u_0 \mid A_{u_0,k})$$

since the distribution of subnetworks of u_0 is indistinguishable across events $C_{u_0,k}$ and $A_{u_0,k}$. By Proposition A.1,

$$\ell(v \mid C_{u_0,k}) \geq \ell(u_0 \mid C_{u_0,k}).$$

For fixed values of k and averaging over all $u_0 \in N(v)$, the $C_{u_0,k}$ are all equally likely, disjoint, and are exhaustive in the event $|M_\sigma| = k$, so we have

$$\ell(v \mid |M_\sigma| = k) = \frac{1}{d} \sum_{u_0 \in N(v)} \ell(v \mid C_{u_0,k}) \geq 1 - \frac{d(d+1)}{kd - k(k-1)/2}\varepsilon$$

Finally, we average over all possible values of $|M_\sigma|$ to get

$$\begin{aligned}
\ell(v) &\geq \frac{1}{d+1} \sum_{k=0}^{d} \ell(v \mid |M_\sigma| = k) \\
&\geq \frac{1}{d+1} \sum_{k=1}^{d} \ell(v \mid |M_\sigma| = k) \\
&\geq \frac{d}{d+1} - \frac{\varepsilon}{d+1} \sum_{k=1}^{d} \frac{d(d+1)}{kd - k(k-1)/2} \\
&= \frac{d}{d+1} - \varepsilon d \sum_{k=1}^{d} \frac{1}{kd - k(k-1)/2} \\
&\geq \frac{d}{d+1} - \varepsilon d \sum_{k=1}^{d} \frac{2}{kd} \\
&\geq 1 - \frac{1}{d+1} - 2\varepsilon \ln d
\end{aligned}$$

as desired. □

Proof of Corollary 2.3. Let $\{v_n\}_n$ be the sequence of vertices $v_n \in G_n$ represented by v. There exists some $\varepsilon = \varepsilon(n)$ and some sequence $\{d_n\}_n$ such that v_n has at least d_n neighbors with learning rate at least $\varepsilon(n)$, and $d_n \to \infty$ and $\varepsilon \to 0$ as $n \to \infty$.

For simplicity, we weaken the bound in Lemma A.2 for v_n as

$$\ell(v_n) \geq 1 - \frac{1}{d} - 2\varepsilon \ln d.$$

This bound works for any value of $d \leq d_n$. Take $d = \min(d_n, e^{1/\sqrt{\varepsilon(n)}})$. Thus we have

$$\ell(v_n) \geq 1 - \frac{1}{d_n} - 4\sqrt{\varepsilon}.$$

Both $\frac{1}{d_n} \to 0$ and $\sqrt{\varepsilon} \to 0$ as $n \to \infty$, so

$$\ell(v_n) \to 1$$

and thus v learns. □

Proof of Lemma 3.2. Fix any ordering σ for which $\sigma(v) = i$. Let $u : \sigma(u) = i+1$ be the vertex at index $i+1$. Denote σ' to be the ordering nearly identical to σ with a single inversion between indices i and $i+1$ that swaps u and v. We note this establishes a bijection between the orderings in A_i and A_{i+1}.

If $u \notin N(v)$, v sees the same neighbors in both orderings. As the neighbors' behaviors are unaffected, v must behave identically under both σ and σ', so $\Pr(a_v = \theta|\sigma) = \Pr(a_v = \theta|\sigma')$. Otherwise, if $u \in N(v)$, then v sees the prediction a_u under σ' not previously visible in σ. By Proposition 2.1, we have that $\Pr(a_v = \theta|\sigma) \leq \Pr(a_v = \theta|\sigma')$. In either case, we sum over all σ and corresponding σ' to get that $\Pr(a_v = \theta|A_i) \leq \Pr(a_v = \theta|A_{i+1})$, as desired. □

B FRAGILE STRATEGIC-ORDER NETWORKS

We find that fixed-order networks are surprisingly fragile to changes in various parameters. Prior work, such as [20] and [4], analyzes networks which achieve learning for all values of q, the parameter which controls the accuracy of private signals. However, here we demonstrate two networks which have significantly different behaviors at different values of q. One learns only when q is high, and one learns only when q is low.

Proof of Proposition 3.1. Consider network $\mathcal{F}$ illustrated in Fig. 1. The network $\mathcal{F} = \{G_n\}_{n\geq 1}$ contains all graphs G_n parametrized by a positive integer n as follows:

G_n contains the vertices $u_0, v, v_1, v_2, v_3, v_4$, the vertices w_i and w'_i for $1 \leq i \leq \log n$, and the vertices u_j for $1 \leq j \leq n-2\log n-6$. There are directed edges (v_i, v) for each v_i, edges (v, w_i) and (w'_i, w_i) for each w_i, and edges (u_{i-1}, u_i) for each u_i where $i \geq 1$.

Then, observe that v_i and w'_i are sources in the directed graph. Since u_0 has a learning rate strictly greater than q, each u_i will ignore its own private signal & follow the signal of u_0 passed down from the agent u_{i-1} before it. Thus, each u_i has the same learning rate as u_0. As $n \to \infty$, the u_i consist of almost all the vertices in the network, implying that $L(G_n) \to \ell(u_0)$ as $n \to \infty$.

We have that agent v learns the correct answer with probability

$$\ell(v,q) = q^5 + 5q^4(1-q) + 10q^3(1-q^2).$$

Recall a_v denotes the prediction that agent v makes, and p_{w_i} denotes the private signal of w_i. Let $S_i = \{p_{w_i}, a_{w'_i}, a_v\}$ be the set of relevant information available to agent w_i. Since its corresponding w'_i always transmits its own private signal, each w_i essentially obtains two private signals. If the signals differ, or both are the same as v, then w_i will predict the same value as v. Otherwise if both signals are opposite v's signal, then w_i infers that the probability v is correct is

$$P(a_v = \theta | S_i) = \frac{f(q)}{f(q) + f(1-q)},$$

where $f(q) = (1-q)^2\ell(v,q)$. There exists a value $q_0 \approx 0.7887$ such that if $q < q_0$, then $P(\theta = a_v) < \frac{1}{2}$ so w_i follows its private signal which is the opposite of a_v, and if $q > q_0$ then $P(\theta = a_v) > \frac{1}{2}$ so w_i predicts a_v.

Suppose $q < q_0$. If $a_v = \theta$, then $a_{w_i} = \theta$ so long as w_i and w'_i aren't both given the incorrect private signal, occurring with independent probability $1-(1-q)^2$. Otherwise if $a_v = 1-\theta$, then $a_{w_i} = \theta$ with probability q^2. Note that $1-(1-q)^2 > q^2$ for all $0 < q < 1$. By the weak law of large numbers, as $n \to \infty$, the fraction of the a_{w_i} which are correct approaches either q^2 or $1-(1-q)^2$ with high probability. Depending on which occurs, u_0 can directly determine the correct ground truth, so $\ell(u_0) \to 1$ and thus $\mathcal{F}$ achieves asymptotic learning.

If $q > q_0$ then $a_{w_i} = a_v$ always, so u_0 only observes the value of a_v. Since $\ell(v) > q$, u_0 follows the observed signal and achieves learning rate $\ell(u) = \ell(v)$. This is constant for arbitrarily large values of n so $\mathcal{F}$ does not achieve asymptotic learning. □

We use the same idea to construct a network that does the opposite, achieving asymptotic learning for high values of q and not for low values of q.

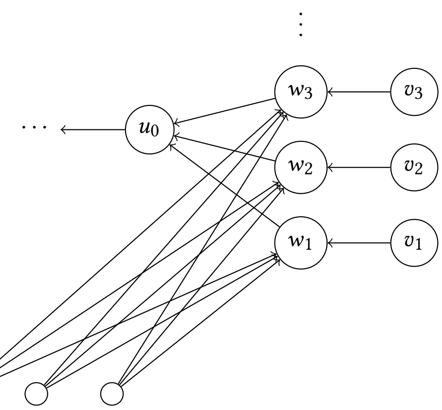

Figure 5: Network that obtains learning for $q > q_0$ but not for $q < q_0$

Proposition B.1. *There exists a fixed-order network $\mathcal{F}$ that obtains asymptotic learning for $q > q_0$ and doesn't learn for $q < q_0$, for some fixed threshold q_0.*

Proof. Consider the network $\mathcal{F}$ shown in Fig. 5, defined as $\mathcal{F} = \{G'_n\}_{n\in\mathbb{N}}$ as follows: G'_n contains the vertices x_1, x_2, x_3, the vertices w_i for $1 \leq i \leq \log n$, the gadgets v_i for $1 \leq i \leq \log n$ consisting of vertices v_{ij} for $0 \leq j \leq 4$, and the vertices u_i for $0 \leq i \leq n - 5\log n - 4$. G'_n contains the edges (x_1, w_i), (x_2, w_i), (x_3, w_i), (v_{i0}, w_i) and (v_{ij}, v_{i0}) for all $1 \leq i \leq \log n$ and $1 \leq j \leq 4$. It also contains the edges (w_i, u_0) for all $1 \leq i \leq \log n$ and (u_{i-1}, u_i) for all $1 \leq i \leq n - 6\text{‘}\log n - 4$. As in the previous graph, note that $L(G'_n) \to \ell(u_0)$ as $n \to \infty$.

We again have that $\ell(v,q) = q^5 + 5q^4(1-q) + 10q^3(1-q^2)$. Let $S_i = \{p_{w_i}, a_{v_i0}, a_{x_1}, a_{x_2}, a_{x_3}\}$ denote the information available to node w_i. As x_1, x_2, x_3 each transmit their own private signal, each w_i functionally gets 4 private signals and the signal of v.

Consider when $q > q_0$ (using the same value of $q_0 \approx 0.7887$). For any $i \in [\log n]$, if the 4 private signals received by w_i are split evenly (two 0s and two 1s) or there is a strict majority in favor of $a_{v_{i0}}$, w_i will follow the prediction of v_{i0}. Otherwise, if there are three predictions for $1 - a_{v_{i0}}$, we have that

$$\Pr(a_{v_{i0}} = \theta | S_i) = \frac{g(q)}{g(q) + g(1-q)} > \frac{1}{2}$$

for $g(q) = q(1-q)^3\ell(v,q)$.

We now case on the predictions of x_1, x_2, x_3. If all 3 are given the correct private signal, $w_i = \theta$ so long as either $p_{w_i} = \theta$ or $a_{v_{i0}} = \theta$, which occurs with probability $q(1-\ell(v_{i0})) + \ell(v_{i0}) \in (q, \ell(v_{i0}))$. If exactly 1 or 2 are correct, $a_{w_i} = a_{v_{i0}}$ unconditionally so $w_i = \theta$ with probability $\ell(v_{i0})$. If all are wrong, $w_i = \theta$ only when $p_{w_i} = \theta$ and $a_{v_{i0}} = \theta$, which occurs with probability $q\ell(v_{i0})$. For any $q > q_0$, these probabilities are all distinct and strictly greater than $1/2$, so as $n \to \infty$, the proportion of w_i predicting the majority vote accounts for $q^* \in (1/2, 1)$ of the w_i's, where $q^* \in \{q(1-\ell(v_{i0})) + \ell(v_{i0}), \ell(v_{i0}), q\ell(v_{i0})\}$. By the weak law of large numbers, u_0 will be able to determine the ground truth with high probability by simply taking the majority of the predictions among the w_i, in which case $\mathcal{F}$ achieves asymptotic learning.

Otherwise, when $q < q_0$, with probability $(1-q)^3$, all x_1, x_2, x_3 predict $1-\theta$. In this case, each w_i follows the signals of these agents regardless of the prediction of v_{i0}. Subsequently, u_0 and the rest of the network follow this signal, which is incorrect with constant probability. Thus, $\mathcal{F}$ does not achieve asymptotic learning. □

C CORRECTNESS OF ALGORITHM 1

Proof of Claim 4.4. By uniformity, any celebrity w_i sees at least $\sqrt{k}$ of its guinea pigs with probability $\geq 1 - 1/\sqrt{k}$. Conditioned on this event, w_i sees at least $1 + \sqrt{k}$ private signals. As per Chpt. 16 of [13], w_i predicts the correct ground truth so long as at least half of the signals are correct, which occurs with probability $\geq 1 - \exp(-(1+\sqrt{k})/(8q^2)) = 1 - \exp(-\Theta(k))$ by a straightforward Chernoff bound. Then by Proposition 2.1, each w_i learns with probability $\geq 1 - O(1/\sqrt{k})$. As $k = \omega(1)$, all w_i's learn. Each $v \in S_n$ contains the vertices $w_1, \ldots w_k$ as neighbors, so the result follows immediately by Corollary 2.3. □

We now provide a simple combinatorial result that will be helpful in many of the following proofs, namely Lemma 4.5:

Lemma C.1. *Given a set $S = \{v_1, \ldots, v_k\}$, $S \subseteq [n]$, let Y be a random variable denoting the number of elements of S with index at most $n\varepsilon$. Then*

$$\Pr(k\varepsilon/2 \leq Y \leq 3k\varepsilon/2) \geq 1 - \frac{4}{k\varepsilon}$$

Proof. Let X_i be an indicator random variable equal to 1 iff v_i arrives before index $n\varepsilon$, and observe that $Y = \sum_{i=1}^k X_i$. By linearity,

$$\mathbb{E}[Y] = \sum_{i=1}^k \mathbb{E}[X_i] = k\varepsilon$$

where we use that $\mathbb{E}[X_i] = \Pr(X_i = 1) = \varepsilon$ for all $i \in [k]$. We now compute

$$\mathrm{Var}(Y) = \sum_{i=1}^k \mathrm{Var}(X_i) + \sum_{i \neq j} \mathrm{Cov}(X_i, X_j)$$

We note $\mathrm{Var}(X_i) \leq \mathbb{E}[X_i^2] = \mathbb{E}[X_i]$ as $X_i \in \{0,1\}$, and thus $\sum_{i=1}^k \mathrm{Var}(X_i) \leq \mathbb{E}[Y]$. Furthermore,

$$\begin{aligned} \mathrm{Cov}(X_i, X_j) &= \mathbb{E}[X_i X_j] - \mathbb{E}[X_i]\mathbb{E}[X_j] \\ &= \frac{n\varepsilon(n\varepsilon - 1)}{n(n-1)} - \varepsilon^2 \\ &\leq 0 \end{aligned}$$

and thus $\mathrm{Var}(Y) \leq \mathbb{E}[Y]$. By Chebyshev's inequality,

$$\Pr(Y < k\varepsilon/2 \cup Y > 3k\varepsilon/2) \leq \Pr(|Y - \mathbb{E}[Y]| \geq k\varepsilon/2) \leq \frac{\mathrm{Var}(Y)}{k^2\varepsilon^2/4} \leq \frac{4}{k\varepsilon}$$

which gives us $\Pr(k\varepsilon/2 \leq Y \leq 3k\varepsilon/2) \geq 1 - \frac{4}{k\varepsilon}$. □

Proof of Lemma 4.5. Let E be the event that v is reachable from the set of vertices in S, and A be the event $\sigma(v) \leq n - \varepsilon\alpha n$ for some unspecified $\alpha = \omega(1), \alpha = o(\frac{1}{\varepsilon})$. We are given that $\Pr(E) = 1 - \delta$, and also have that $\Pr(A) = 1 - \alpha\varepsilon$. Then,

$$\Pr(E) = \Pr(E \cap A) + \Pr(E \cap A^c) \leq \Pr(E \cap A) + \alpha\varepsilon$$

$$\therefore \Pr(E \cap A) \geq 1 - \delta - \alpha\varepsilon$$

We now compute the probability v is reached by some vertex in S after index $n\varepsilon\alpha$. Let $\sigma[x : y]$ denote the subgraph induced by vertices between indices $x, y \in [n]$ in the ordering, and let E' be the event that v is reachable by some $u \in S$ where $\sigma(u) \geq n\varepsilon\alpha$. We claim $\Pr(E') \geq \Pr(E \cap A)$. Indeed, for any ordering $\sigma \in E \cap A$, observe we may bijectively map each σ to a new ordering $\sigma' = \sigma[n - n\varepsilon\alpha + 1 : n] \circ [1 : n - n\varepsilon\alpha]$, i.e. we move the suffix of length $n\varepsilon\alpha$ to the front of the ordering. Note that v is reachable by some $u \in \sigma[1 : n - n\varepsilon\alpha]$, and in σ', $\sigma'(u) \geq n\varepsilon\alpha$. Thus, $\sigma' \in E'$, and as this holds for all $\sigma \in E \cap A$, we conclude $\Pr(E') \geq 1 - \delta - \alpha\varepsilon$.

By Proposition A.1, we have that $\ell(v|E') \geq \ell(u|E')$, where $u \in S, \sigma(u) \geq n\varepsilon\alpha$ is a vertex in v's subnetwork. It remains for us to prove $\ell(u|E')$ is large by showing that, conditioned on $\sigma(u) \geq n\varepsilon\alpha$, at least one of the celebrities w_i added in BoostAgents arrives before u and still learns.

For each $i \in [1/\varepsilon]$, consider the celebrity w_i, and let B_i be the event that w_i appears before index $n\varepsilon\alpha$. Let $X = \sum_i B_i$ denote the total number of celebrities before index $n\varepsilon\alpha$. By Lemma C.1, $X \in [\frac{\alpha}{2}, \frac{3\alpha}{2}]$ with probability at least $1 - \frac{4}{\alpha}$. We note conditioning on event E' only increases the probability of B_i, as E' only concerns indices after $n\varepsilon\alpha$.

For any w_i appearing before index $n\varepsilon\alpha$, let C_i denote the event that it sees at least $\sqrt{\alpha}$ of its guinea pigs. By Proposition 2.1, the learning rate of w_i conditioned on C_i is at least as good as

$$\ell(w_i|C_i) \geq (1 - \exp(-\frac{\sqrt{\alpha}}{8q^2})),$$

using our previously described Chernoff bound. Thus it suffices to show $\Pr(C_i)$ is large. Again applying Lemma C.1, we have probability at least $1 - \frac{4}{\alpha}$ that at least $\alpha/2$ of w_i's guinea pigs appear before index $n\varepsilon\alpha$. Taking a uniformly random ordering, all orderings of w_i with relation to the other guinea pigs before index $n\varepsilon\alpha$ have equal probability, so w_i has probability at least $1 - \frac{2}{\sqrt{\alpha}}$ of seeing at least $\sqrt{\alpha}$ guinea pigs. Thus, $\Pr(C_i|B_i) \geq 1 - \frac{4}{\alpha} - \frac{2}{\sqrt{\alpha}} \geq 1 - \frac{4}{\sqrt{\alpha}}$ for sufficiently large α.

We now have that

$$\begin{aligned} \ell(u|E') &\geq \ell(u|E', \exists\, i : B_i = 1) \Pr(\exists\, i : B_i = 1|E') \\ &\geq \ell(w_i|E', B_i = 1)\left(1 - \frac{4}{\alpha}\right) \\ &\geq \ell(w_i|B_i = 1, C_i) \Pr(C_i|B_i)\left(1 - \frac{4}{\alpha}\right) \\ &\geq \left(1 - e^{-\sqrt{\alpha}/(8q^2)}\right)\left(1 - \frac{4}{\sqrt{\alpha}}\right)\left(1 - \frac{4}{\alpha}\right) \\ &\geq 1 - \frac{9}{\sqrt{\alpha}} \end{aligned}$$

for $\alpha = \omega(1)$. Finally, we have that

$$\ell(v) \geq \ell(v|E') \Pr(E') \geq (1 - 9/\sqrt{\alpha})(1 - \delta - \alpha\varepsilon) \geq 1 - \delta - 2\sqrt[3]{81\varepsilon}$$

where our desired inequality is obtained by setting $\alpha = \sqrt[3]{\frac{81}{\varepsilon^2}}$. □

Proof of Lemma 4.6. Consider any modification strategy using at most $M := M(n)$ modifications to $G := G_n$, $G = \{V, E\}$, where the resulting modified graph G_M obtains learning. Let $S \subseteq V$ be the set of all vertices u that are involved in a modification (i.e., an edge incident on u is added/deleted). We show that calling BoostAgents(G, S, k) on G to obtain the modified graph G_B results in a network that learns, for any superconstant $k = \omega(1)$.

Consider any vertex v not originally a learner under G that learns in G_M. With some probability, v's subnetwork must contain some vertex $u \in S$ involved in a modification. Let A be the event v is reachable by any such u, and denote $\ell^M(v)$ to be the random-order learning rate of v in G_M. Then,

$$\ell^M(v) = \ell^M(v|A)\Pr(A) + \ell^M(v|A^c)\Pr(A^c) = 1 - o(1)$$

We now show the learning rate of v in G_B, $\ell(v)$, is at most a $o(1)$ additive correction less than $\ell^M(v)$. In both G_M and G_B, conditioned on A^c, the subnetwork of v is identical to that in G, and thus the learning rate remains unchanged. Furthermore, $\Pr(A^c)$ depends only on the relative ordering of the original vertices in V and is hence identical in both graphs. Now, let A' denote the probability that there exists a modified vertex $u \in S$ within v's subnetwork appearing after index $n\varepsilon\alpha$, where we may set $\alpha = 1/\sqrt{\varepsilon}$. By the analysis of Lemma 4.5, $\Pr(A') \geq \Pr(A) - \sqrt{\varepsilon}$, and $\ell(v|A') \geq 1 - 5\varepsilon^{1/4} \geq \ell^M(v) - o(1)$. Thus,

$$\begin{aligned}
\ell(v) &= \ell(v|A)\Pr(A) + \ell(v|A^c)\Pr(A^c) \\
&\geq \ell(v|A')\Pr(A') + \ell(v|A^c)\Pr(A^c) \qquad (A' \subseteq A) \\
&\geq (\ell^M(v|A) - o(1))(\Pr(A) - o(1)) + \ell^M(v|A^c)\Pr(A^c) \\
&\geq \ell^M(v) - o(1)
\end{aligned}$$

and thus v learns in G_B. As this holds for all viable non-learning $v \in G$ and since G_M learns, we conclude that G_B learns as well.

Let ALG be the number of edges/vertices added to construct G_B. Note that BoostAgents adds at most $2k$ edges for each edge modification in G_M, along with k^2 additional edges and $k^2 + k$ vertices. This gives us $ALG \leq k \cdot (2M + 1) + 2k^2 = O(k) \cdot M + 2k^2$, as desired. ◻

Proof of Claim 4.7. Nonnegativity and monotonicity are inherent to reachability: that is, $C_\sigma(T) \geq C_\sigma(S) \geq 0$ for any $T \supseteq S$. On submodularity, we fix σ and consider some agent v. For any sets $S \subseteq T \subseteq V'$, we have that $C_\sigma(S) \subseteq C_\sigma(T)$. As such, $C_\sigma(\{v\}) \setminus C_\sigma(S) \supseteq C_\sigma(\{v\}) \setminus C_\sigma(T)$. But $\frac{1}{n!}\sum_\sigma |C_\sigma(\{v\}) \setminus C_\sigma(S)| = f(S \cup \{v\}) - f(S)$ and $\frac{1}{n!}\sum_\sigma |C_\sigma(\{v\}) \setminus C_\sigma(T)| = f(T \cup \{v\}) - f(T)$, so we are done. ◻

Proof of Lemma 4.8. At any iteration, let $\tilde{c}_v = \tilde{C}(S \cup \{v\}) - \tilde{C}(S)$. By Monte Carlo guarantees, our estimate $\tilde{C}(S \cup \{v\})$ is within an additive ε of the true value $C(S \cup \{v\})$ with probability $\geq 1 - \delta = 1 - \frac{1}{n^9}$. Across all n iterations, the probability that no estimate falls outside of the error is $1 - \frac{1}{n^8}$.

Even so, our algorithm may pick a suboptimal v at each iteration. Let $S^{(i)}$ be the set of size i that a exponential time greedy algorithm with exact knowledge of $C(S)$ would obtain, and let $S^{(i)}_{ALG}$ be the set of size i obtained by our algorithm (equivalently, this is the set obtained after i iterations of the main loop). As we incur $\leq \varepsilon = \frac{1}{n^2}$ error at each iteration, we have for all $i \leq n$,

$$C(S^{(i)}) \leq C(S^{(i)}_{ALG}) + i\varepsilon \leq C(S^{(i)}_{ALG}) + \frac{1}{n}$$

Let S be the final set obtained by the exponential time greedy algorithm, covering $\geq |V'| - T$ vertices, and S_{ALG} be our algorithm's final boosting set. Suppose $|S| = j$, i.e. after j iterations, the exponential time greedy algorithm has $C(S^{(j)}) \geq |V'| - T$. Since each uncovered vertex v has probability $\frac{1}{1+\deg v} \geq \frac{1}{2n}$ of being first among its neighbors & unreachable from S, each iteration increments $\tilde{C}(S^{(i)}_{ALG})$ by at least $\frac{1}{2n}$. We conclude that $C(S^{j+2}_{ALG}) \geq |V'| - T$ with high probability. Hence, the final set S_{ALG} obtained by our randomized algorithm is at most 2 larger than the set S obtained if we could compute C exactly. For the rest of our analysis, we now bound the size of S, knowing $S_{ALG} \leq (1 + \frac{2}{|S|})S$.

Let S^* be the optimal boosted set. It remains for us to show that $|S| \leq |S^*| \cdot O(\log n)$, which directly implies $ALG \leq M_B \cdot O(\log n)$ for any M_B. This would be sufficient to obtain $ALG \leq M \cdot O(k \log n) + 2k^2$ as the boosting construction using M_B modifications in Lemma 4.6 adds at most $2k$ edges for each of the M modifications made under an arbitrary strategy. For sufficiently small $k = O(\log n)$, the additive $O(k^2)$ term can be merged with the $M \cdot O(k \log n)$ term, giving us the desired result.

Let $u_i = |V'| - C(S)$ denote the number of non-learning vertices still uncovered by Algorithm 1 when $|S| = i$. The exponential time algorithm terminates when $u_i \leq T$. Note that $|V'| - C(|S^*|) \leq T$; consequently, at any point in the algorithm, there exists a sequence of at most M_B vertices that we may add to S to cover all but at most T non-learners, as $|V'| - C(S \cup S^*) \leq T$. By submodularity of coverage (Claim 4.7), we may invoke the greedy approximation guarantee in [19] to obtain

$$u_{i+M_B} \leq \frac{u_i}{e}$$

for any iteration $i \geq 0$. As $u_0 = |V'|$, after i iterations,

$$u_i \leq |V'| \left(\frac{1}{e}\right)^{\lfloor \frac{i}{M_B} \rfloor}$$

The algorithm terminates when $u_i < T$, so the number of iterations of the main loop is $i \leq M_B \cdot \ln \frac{|V'|}{T} \leq M_B \cdot O(\log n)$. The number of iterations is precisely the size of S, and thus $|S| \leq O(\log n)|S^*|$. ◻

Proof of Theorem 4.9. By Lemma 4.8, Algorithm 1 $O(k \log n)$-approximates the number of modifications needed to achieve learning with high probability. Taking $k = g(n)$ for any $g(n) = \omega(1)$, $g(n) = O(\log n)$, we obtain the desired approximation guarantee. Furthermore, Algorithm 1 outputs a set S covering at least $|V'| - T$ vertices w.p. $\geq 1 - \frac{1}{n^8}$. By Lemma 4.5, with probability approaching 1, all but $o(n)$ vertices are boosted to learn, so the algorithm output is indeed a learning network.

It remains for us to analyze the runtime of Algorithm 1. Sampling N random orderings takes $O(n^7 \log n)$, assuming each ordering can be generated in $O(n)$ time. For each ordering, computing $\tilde{c}_v = \tilde{C}(S \cup \{v\}) - \tilde{C}(S)$ can be done with a single BFS per ordering, which takes $O((m+n)n^6 \log n)$ time per vertex and thus $O((m+n)n^7 \log n)$ time in aggregate. The runtime of BoostAgents is $O(kn) = O(n^2)$, so the overall runtime is $O((m+n)n^7 \log n)$, which is polynomial. ◻

D ADDITIONAL PROOFS IN SECTION 4

Proof of Proposition 4.1. We will prove the contrapositive. Let $\alpha(G)$ denote the maximum independent set size of G, and let the set of agents guaranteed to predict their private signal be $S \subseteq V$. We note that any agent $v \in V \setminus S$ will follow its neighbors if they unanimously agree on some binary value. However, the only agents guaranteed to predict their private signals are sources (in-degree 0), and potentially some agents with in-degree 1 who predict their private signal due to the tie-breaking rule. For each vertex v, this occurs only if v is one of the first two vertices in the ordering among $\{v\} \cup N(v)$. A classical result (first shown in [9] and [31]) states that

$$\alpha(G) \geq \sum_{v \in V} \frac{1}{1 + \deg(v)},$$

so the expected number of agents with in-degree 0 or 1 is at most $\sum_{v \in V} \frac{2}{1+\deg(v)} \leq 2\alpha(G)$. Thus with probability at least 1/2 over the random ordering, $|S| \leq 4\alpha(G)$ by Markov's inequality. When $\alpha(G) = O(1)$, we observe the probability of all agents in S being given the wrong signal (and hence predicting the wrong binary value) is at least $(1-q)^{4\alpha(G)} = \Omega(1)$. Should this occur, we inductively observe that all vertices in $V \setminus S$ will herd to the wrong binary value. Thus, with fixed probability, the entire graph herds to the incorrect binary value and therefore doesn't learn. □

Proof of 4.2. For any sufficiently large n, our construction leverages any functions $h(n), g(n) = \omega(1)$ for which $h(n) \cdot g(n) = o(n)$.

We first take the complete graph K_n and select $g(n)$ special vertices, denoted by the set S. Equip each special vertex with $h(n)$ guinea pigs (vertices with degree 1). Let G_m be the resulting graph, for $m = n + h(n)g(n)$. Without the vertices of degree 1, we are left with K_n, which doesn't achieve asymptotic learning under any ordering, as discussed in [7].

To show the network $\mathcal{F} = \{G_m\}$ learns, it suffices to show that the vertices within the K_n subgraph learn, as $o(m)$ vertices fall outside this subgraph. For $u \in S$, let A be the event that u sees at least $\sqrt{h(n)}$ many of its guinea pigs. Over the uniformly random ordering, $\Pr(A) \geq 1 - \frac{1}{\sqrt{h(n)}}$. As long as at least half of its guinea pigs predict the correct ground truth, so too will u (see Chpt. 16 of [13]). By a Chernoff bound and Proposition 2.1, conditioned on seeing at least $\sqrt{h(n)}$ many independent private signals, we have that

$$\ell(u|A) \geq 1 - \exp\left(-\frac{\sqrt{h(n)}}{8q^2}\right)$$

so by Lemma 2.2, we have

$$\ell(u) \geq \left(1 - \frac{1}{\sqrt{h(n)}}\right)\left(1 - e^{-\sqrt{h(n)}/(8q^2)}\right).$$

For $h(n) = \omega(1)$, we conclude that each special vertex learns.

Since all vertices in the K_n subgraph have at least $g(n) = \omega(1)$ learning neighbors, by Corollary 2.3, all such vertices achieve learning as well. We conclude that $\mathcal{F}$ achieves asymptotic learning. □

Proof of Proposition 4.3. Within any graph G_m (constructed as per the proposition statement), let $V^{(i)}$ be the vertex set of the ith copy of the graph G'_k, and $v_k^{(i)} \in V^{(i)}$ be the corresponding vertex in $V^{(i)} \cap S$. Recall $k = \log\log n$. We fix an index $j = n/k$, and for each $i \in \sqrt{n}$, let A_i be the indicator random variable equal to 1 iff all vertices in $V^{(i)}$ appear at/before index j. By uniformity over the ordering, we compute

$$\begin{aligned}
\Pr(A_i = 1) &= \frac{\prod_{l=0}^{k-1}(j-l)}{\prod_{l=0}^{k-1}(n-l)} = \prod_{l=0}^{k-1} \frac{j-l}{m-l} \\
&\geq \left(\frac{j-k}{m-k}\right)^k \\
&\geq \left(\frac{1}{2\log\log n}\right)^{\log\log n} \\
&\geq \frac{1}{\log n (\log\log n)^{\log\log n}} \\
&\geq \frac{1}{n^\delta}
\end{aligned}$$

for any $\delta > 0$. Let X be a random variable denoting the number of graphs G'_k that appear at/before index j. As $X = \sum_{i \in [\sqrt{n}]} A_i$, by linearity, $\mathbb{E}[X] \geq n^{1/2-\delta}$. Furthermore, by linearity of variance,

$$\mathrm{Var}(X) = \sum_{i \in [\sqrt{n}]} \mathrm{Var}(A_i) + \sum_{i \neq l} \mathrm{Cov}(A_i, A_l)$$

We first note $\mathrm{Var}(A_i) \leq \mathbb{E}[A_i^2] = \mathbb{E}[A_i]$ as $A_i \in \{0,1\}$. We also observe $\mathrm{Cov}(A_i, A_l) \leq 0$, as

$$\begin{aligned}
&\mathrm{Cov}(A_i, A_l) = \mathbb{E}[A_i A_l] - \mathbb{E}[A_i]\mathbb{E}[A_l] \\
=&\frac{\prod_{l=0}^{2k-1}(j-l)}{\prod_{l=0}^{2k-1}(m-l)} - \left(\frac{\prod_{l=0}^{k-1}(j-l)}{\prod_{l=0}^{k-1}(m-l)}\right)^2 \\
\leq&\left(\frac{\prod_{l=0}^{k-1}(j-l)}{\prod_{l=0}^{k-1}(m-l)}\right)^2 - \left(\frac{\prod_{l=0}^{k-1}(j-l)}{\prod_{l=0}^{k-1}(m-l)}\right)^2 \\
=&0
\end{aligned}$$

where the penultimate line uses that

$$\frac{\prod_{l=k}^{2k-1}(j-l)}{\prod_{l=k}^{2k-1}(m-l)} \leq \frac{\prod_{l=0}^{k-1}(j-l)}{\prod_{l=0}^{2k-1}(m-l)}$$

so $\mathrm{Var}(X) \leq \mathbb{E}[X]$. Hence, by Chebyshev's inequality,

$$\Pr(|X - \mathbb{E}[X]| \geq \mathbb{E}[X]/2) \leq \frac{\mathbb{E}[X]}{\mathbb{E}[X]^2/4} \leq \frac{4}{n^{1/2-\delta}}$$

so we conclude that with probability $\geq 1 - \frac{4}{n^{1/2-\delta}}$ that at least $n^{1/2-\delta}/2$ sets $V^{(i)}$ appear entirely before index j.

Without loss of generality, let $S' = [1, \ldots, L]$ be the indices corresponding to sets appearing before index j. Denote B_i to be the event that the relative ordering on $V^{(i)}$ is identical to σ_k. As the vertex sets $V^{(i)}$ are disjoint, the events B_i for $i \in S'$ are mutually independent with each other and with the events A_i. Thus,

$$\begin{aligned}
\Pr\left(\bigcap_{i \in S'} \overline{B_i} \mid |S'| \geq n^{1/2-\delta}/2\right) &= \prod_{i \in S'} \Pr\left(\overline{B_i} \mid |S'| \geq n^{1/2-\delta}\right) \\
&= \prod_{i \in S'} \Pr(\overline{B_i}|A_i) = \prod_{i \in S'} \Pr(\overline{B_i}) \\
&= (1 - 1/k!)^L
\end{aligned}$$

where the last line uses that $\Pr(B_i) = 1/k!$. When $L \geq n^{1/2-\delta}/2$, since $1/k! \leq 1/(\log\log n)^{\log\log n} \leq 1/n^{\delta}$, we use that $1 - x \leq e^{-x}$ to get that $(1 - 1/k!)^L \leq \exp(-n^{1/2-2\delta}/2)$. Thus,

$$\Pr\left(\bigcup_{i\in S'} B_i \mid S'\right) = 1 - \Pr\left(\bigcap_{i\in S'} \overline{B_i} \mid S'\right) \geq 1 - \exp(-n^{1/2-2\delta}/2)$$

Finally, to show our network learns, consider any vertex u in the subgraph $u \in K_n$. Let $\ell_{\sigma_k}(v_k) = 1 - \varepsilon = 1 - o(1)$, and note that this lower bounds $\ell(v_k^{(i)}|A_i = 1)$ by Proposition 2.1. Let $\ell(u|j)$ be the learning rate of u conditioned on being at index j. By Lemma 3.2, $\ell(u) \geq \ell(u|j)\Pr(\sigma(u) \geq j) = \ell(u|j)(1 - \frac{1}{\log\log n})$, it remains to show $\ell(u|j)$ goes to 1 as $n \to \infty$.

With probability at least $\Pr(|S'| \geq n^{1/2-\delta}, \bigcup_{i\in S'} B_i)$, any u at vertex j sees at least one $v_k^{(i)}$ that learns with probability $1 - \varepsilon$. Thus,

$$\ell(u|j) \geq (1-\varepsilon)\left(1 - \exp\left(\frac{-n^{1/2-2\delta}}{2}\right)\right)\left(1 - \frac{4}{n^{1/2-\delta}}\right) = 1 - o(1)$$

for any $0 < \delta < 1/4$. We conclude that each $u \in K_n$ learns, and as the K_n subgraph consists of all but $o(m)$ vertices, the network $\mathcal{F} = \{G_m\}$ obtains asymptotic random-order learning. □

Proof of Claim 4.11. Fix n and let $v := v_n$. Let $m = \sum_{u\in N(v)} \frac{1}{\deg u} = \omega(1)$, $N'(v) = \{u \in N(v) : \deg u \leq \deg v/\sqrt{m}\}$, and $d = |N'(v)|$. Note that $d > m$, and

$$\begin{aligned}\sum_{u\in N(v_n)} \frac{1}{\deg u} &= \sum_{u\in N'(v)} \frac{1}{\deg u} + \sum_{u\notin N'(v)} \frac{1}{\deg u}\\ &\leq \sum_{u\in N'(v)} \frac{1}{\deg u} + \frac{\deg v}{\deg v/\sqrt{m}}\\ &= \sum_{u\in N'(v)} \frac{1}{\deg u} + \sqrt{m}\end{aligned}$$

Thus, $\sum_{u\in N'(v)} \frac{1}{\deg u} \geq m - \sqrt{m} \geq m/2$.

Observe that a superconstant number of vertices in $N'(v)$ appear very early in the ordering. In particular, let X be the number of vertices in $N'(v)$ arriving before index $\frac{nl}{d}$. By Lemma C.1, $l/2 \leq X \leq 3l/2$ with probability at least $1 - \frac{4}{l}$.

Let $S \subseteq N'(v)$ be the set of vertices appearing before index $\frac{nl}{d}$. We now show that with high probability, all vertices in S are the first among their neighbors. For each $u \in S$, let B_u be the event that u is the first among its neighbors. For each $j \in [\deg u]$, let B_{uj} be the event that neighbor j is after index $\frac{nl}{d}$. Then $\bigcap_j B_{uj} \cap \{u \in S\} \subseteq B_u \cap \{u \in S\}$, and

$$\Pr\left(\bigcup_j \overline{B_{uj}} \mid u \in S\right) \leq \frac{l\deg u}{d} \leq \frac{l}{\sqrt{m}}$$

by a union bound, where $\Pr(\overline{B_{uj}}|u \in S) \leq \frac{l}{d}$. Thus, $\Pr(B_u|u \in S) \geq 1 - \Pr(\bigcup_j \overline{B_{uj}}|u \in S) \geq 1 - l/\sqrt{m}$. We conclude that $\Pr(\overline{B_u}|u \in S) \leq l/\sqrt{m}$, and thus

$$\begin{aligned}\Pr\left(|S| \in \left[\frac{l}{2}, \frac{3l}{2}\right], \bigcap_{u\in S} B_u\right) &= \Pr\left(\bigcap_{u\in S} B_u \mid S\right)\Pr\left(|S| \in \left[\frac{l}{2}, \frac{3l}{2}\right]\right)\\ &\geq \left(1 - \Pr\left(\bigcup_{u\in S} \overline{B_u} \mid S\right)\right)\left(1 - \frac{4}{l}\right)\\ &\geq \left(1 - \frac{3l^2}{2\sqrt{m}}\right)\left(1 - \frac{4}{l}\right)\end{aligned}$$

where the last line uses a union bound to obtain $\Pr(\bigcup_{u\in S} \overline{B_u}|S) \leq \frac{3l}{2}\Pr(\overline{B_u}|u \in S)$. We may set $l = m^{1/6} = \omega(1)$ to get that, with probability $1 - \Theta(m^{-1/6}) = 1 - o(1)$, $|S| \in [l/2, 3l/2]$ and each $u \in S$ is the first among its neighbors. Let the intersection of both these events be denoted E.

To conclude, we show the learning rate of our vertex v conditioned on being at index nl/d is high. As $\ell(v|i) \geq \ell(v|nl/d)$ for any $i \geq nl/d$, by Lemma 3.2, we note $\ell(v) \geq \ell(v|\frac{nl}{d})\Pr(\sigma(v) > \frac{nl}{d}) \geq \ell(v|\frac{nl}{d})(1 - \frac{l}{d})$. As $1 - \frac{l}{d} = 1 - o(1)$, it remains to show $\ell(v|\frac{nl}{d})$ goes to 1 as $n \to \infty$. We have that

$$\begin{aligned}\ell\left(v \mid \frac{nl}{d}\right) &\geq \ell\left(v \mid \frac{nl}{d}, E\right)\Pr\left(E \mid \sigma(v) = \frac{nl}{d}\right)\\ &\geq \ell\left(v \mid \frac{nl}{d}, E\right)(1 - \Theta(m^{-1/6}))\end{aligned}$$

Conditioned on E and being after index nl/d, vertex v sees at least $m^{1/6}/2$ neighbors who arrive first in their neighborhood, who simply predict their private signal. By a Chernoff bound, the learning rate obtained by taking the majority of these private signals is $\geq 1 - \exp(-m^{1/6}/(16q^2))$, so by Proposition 2.1, $\ell(v|\frac{nl}{d}, E) \geq 1 - \exp(-\Theta(m^{1/6}))$. We conclude that v learns. □